

1. Administrative & Study Identification

Full Study Title: Extension Study to Assess Effects of Non-interrupted Versus Interrupted and Long Term Treatment of Two Dose Regimes of Secukinumab in Subjects With Hidradenitis Suppurativa **Short Title/Acronym:** SUNSHINE and SUNRISE Extension Trial **Protocol Number:** CAIN457M2301E1 **NCT Number:** NCT04179175 **Posting ID:** 527867318624633521 **Trial Status:** Active, Not Recruiting **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Secukinumab (Cosentyx), an IL-17A monoclonal antibody (ATC Code: L04AC10) **Phase of Development:** Phase III **Medical Monitor:** Novartis Pharmaceuticals / Central Trial Contact

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the maintenance of Hidradenitis Suppurativa Clinical Response (HiSCR response) at Week 104 in either continuous or interrupted therapy compared to placebo. **Secondary Objectives:** To assess the long-term safety and tolerability evaluated by adverse events, abnormal laboratory values, and vital signs.

2.2 Background & Rationale To assess long-term efficacy, safety, and tolerability of Secukinumab in subjects with moderate to severe hidradenitis suppurativa completing either of the 2 core Phase III studies (CAIN457M2301 / SUNSHINE and CAIN457M2302 / SUNRISE). Because hidradenitis suppurativa is a chronic, relapsing condition, this extension trial with a randomized withdrawal period was designed to determine whether continuous maintenance therapy is superior to treatment interruption, and to characterize the time to Loss of Response (LOR).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Extension Study) **Control Type:** Placebo-controlled (Randomized withdrawal design) **Blinding:** Double-blind **Randomization:** 2:1 ratio; subjects who were HiSCR responders after 52 weeks in the core studies were randomized to either continue their assigned Secukinumab dose regimen for another 52 weeks or be placed on placebo.

3.2 Planned Enrollment & Duration Target \$N\$: 703 participants **Number of Sites:** 191 locations globally **Estimated Study Start:** 18-Mar-2020 **Estimated Primary Completion:** 26-May-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults (aged ≥ 18 years) diagnosed with moderate-to-severe Hidradenitis Suppurativa (HS) for ≥ 1 year. 2. Patients who received treatment through Week 52 and successfully completed either of the core Phase III studies CAIN457M2301 (SUNSHINE) or CAIN457M2302 (SUNRISE).

4.2 Exclusion Criteria 1. Protocol deviation in the core study which would prevent the meaningful analysis of the extension study. 2. Ongoing or planned use of prohibited HS or non-HS treatments. 3. Participation in the extension could expose the subject to an undue safety risk. 4. Current severe progressive or uncontrolled disease rendering the subject unsuitable for continued study participation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Secukinumab 300 mg administered either every 2 weeks (Q2W) or every 4 weeks (Q4W). **Route of Administration:** Subcutaneous (SC) injection **Duration of Treatment:** 52 weeks in the double-blind randomized withdrawal period (up to Week 104 of cumulative treatment). Subjects who lost response (LOR) or were non-responders at Week 52 were transferred to an open-label treatment continuing up to Week 260.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of selected antibiotics permitted during core trials. **Prohibited:** Use of TNF- α inhibitors or other prohibited biologic systemic immunomodulatory therapies during the trial period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	N/A	Rollover transition from Week 52 of core Phase III studies
Baseline	Day 0 (Ext)	Randomization (2:1) into continuous or interrupted therapy groups
Interim	Q2W / Q4W	Subcutaneous injection, HiSCR Evaluation, Safety Labs, AE monitoring
End of Study	Week 104 / 260	Final clinical assessment (LOR determination), Exit Interview
Follow-up	+ 8 Weeks	Post-treatment safety follow-up and final vital signs

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definitions: 1. Time to Loss of Response (LOR) up to Week 104 in Hidradenitis Suppurativa Clinical Response (HiSCR) Responders. 2. Incidence Rate of Participants Achieving Loss of Response (LOR) up to Week 104 in HiSCR Responders. **Measurement Method:** Clinical assessment of

Hidradenitis Suppurativa Clinical Response (HiSCR), measuring the increase in abscesses and inflammatory lesions over time.

7.2 Secondary & Exploratory Endpoints - Number of Participants With Treatment Emergent Adverse Events. - Long-term safety and tolerability evaluated by adverse events (AEs), abnormal laboratory values, and vital signs up to Week 260.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of clinical safety, laboratory abnormalities, and immunogenicity (anti-drug antibodies) at study visits. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols utilized across all 191 trial sites. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee governing patient safety and randomized withdrawal outcomes.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Full Analysis Set (FAS) for efficacy and Safety Set for AE/SAE evaluations. **Sample Size Justification:** Cohort sizes derived sequentially from the completers of the SUNSHINE and SUNRISE core trials ($N = 703$). **Handling of Missing Data:** For the LOR analysis, two separate stratified Cox models were fitted (for Q2W and Q4W). A 2.5% alpha level (one-sided) was utilized to control for the familywise type I error.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Rigorous onsite and remote clinical monitoring across 191 global locations. **Audit Trail:** Standardized tracking of Protocol Deviations (PDs), eCRF locking, and clinical pathway adherence.

11. Study Identifiers & Governance

Primary ID: CAIN457M2301E1 **Registry Number:** NCT04179175 **Posting ID:** 527867318624633521 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** SUNSHINE and SUNRISE Extension Trial **Official Regulatory Title:** A Multicenter, Double-blind, Randomized Withdrawal Extension Study of Subcutaneous Secukinumab to Demonstrate Long-term Efficacy, Safety and Tolerability in Subjects With Moderate to Severe Hidradenitis Suppurativa

12. Clinical Framework (Hidradenitis Suppurativa Specific)

Primary Condition: Hidradenitis Suppurativa (HS) **Disease Codes:** MeSH: D017497 | HPO: HP:0040154 | SNOMED CT: 59393003 **Diagnosis Threshold:** Moderate to severe HS (Hurley stage II or III) with a history of ≥ 1 year **Medical Methodology:** Interleukin-17A (IL-17A) Targeted Monoclonal Antibody Therapy **Optimization Target:** Maintenance of HiSCR (Hidradenitis Suppurativa Clinical Response) and prevention of Loss of Response (LOR)

13. Study Procedures & Methodology

Management Pathway: Continuous vs. Interrupted Biologic Maintenance Therapy **Education Protocol (HCP):** Site Initiation Visits (SIV) and comprehensive training on defining and assessing HiSCR and LOR **Education Protocol (Patient):** Subcutaneous injection technique training and self-reporting of flare-ups **Quality Audit Mechanism:** Independent medical review of lesion counts, centralized AE tracking, and immunogenicity assay reviews

14. Observation & Follow-Up Schedule

Total Duration: Up to 260 weeks (plus 8 weeks post-treatment follow-up) **Onsite Visit Cadence:** Depending on dosing regimen (every 2 weeks [Q2W] or every 4 weeks [Q4W]) **Remote Monitoring Frequency:** Routine safety check-ins and laboratory assessments **Checklist Review Interval:** Continuous clinical evaluation during dosing visits to confirm HiSCR maintenance

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				